

# MIS: Difficulty Scoring as a Pre-Augmentation Diagnostic for Agricultural Object Detection

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**One-sentence claim.** MIS provides structural diagnostics — computable from training-set inference alone — that are consistent with observed copy-paste augmentation failure and that identify the failure conditions before augmentation is conducted.

**Why the claim changed from the original draft.** The original aim was to show that MIS-guided copy-paste improves AP. Three datasets (MinneApple, GWHD, VisDrone) showed uniform failure. The original positive result on MinneApple ( $\Delta\text{AP} = +0.013$  for Score CP vs. Random CP) was an artifact of a contaminated validation split and does not hold on clean data. Rather than search for a positive result, the research is reframed: the consistent failure is itself informative, and MIS correctly identifies the structural reasons for that failure.

## Methodology

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### 1.1 MIS Scoring

Predictions are generated at low confidence threshold  $\tau_{\text{conf}} = 0.001$  (to capture near-miss detections) and matched to ground truth at  $\tau_{\text{IoU}} = 0.5$ . Let  $\mathcal{P}(g)$  be matched predictions for object  $g$ . The **object difficulty score** is:

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g)} [\alpha(1 - c_p) + \beta(1 - u_p)], & \mathcal{P}(g) \neq \emptyset \\ \alpha + \beta = 1.0, & \text{missed object} \end{cases} \quad (1)$$

with  $\alpha = \beta = 0.5$ ,  $c_p = \text{conf}(p)$ ,  $u_p = \text{IoU}(p, g)$ . Taking the minimum is necessary because the low  $\tau_{\text{conf}}$  generates many candidate boxes; minimum selects the best-matching one.

The **image difficulty score** is:

$$S_{\text{img}}(I) = w_1 \cdot \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\text{obj}}(g) + w_2 \cdot \text{FP rate}(I) \quad (2)$$

$w_1 = 1$  (fixed).  $w_2$  is calibrated automatically by grid search to minimize  $|r(S_{\text{img}}, \text{miss\_rate}) - r(S_{\text{img}}, \text{FP\_rate})|$ , balancing both error signals. FP rate is computed at the optimal operating threshold found from validation.

### 1.2 Dataset Splits

**MinneApple.** The dataset contains images from only 10 unique apple trees. A random image-level split causes images from the same tree to appear in both training and validation, inflating validation AP through tree-specific memorization (initial contaminated baseline: AP= 0.439, discarded). We use a **tree-identity-clean split**: validation uses 30 images from trees not in training ( $\approx 3$  images per held-out tree). Test set is the official MinneApple test split. Training: 200 epochs max, early stopping patience 50 epochs.

**MinneApple size thresholds.** COCO thresholds (small  $\leq 32$  px, medium  $\leq 96$  px) do not suit MinneApple at  $720 \times 1280$  resolution. We calibrate to the actual training distribution:

| Category   | Criterion                                                                 | Train % |
|------------|---------------------------------------------------------------------------|---------|
| Very Small | area $\leq 400 \text{ px}^2$ ( $< 20 \times 20 \text{ px}$ )              | 23.08%  |
| Small      | $400 < \text{area} \leq 1024 \text{ px}^2$ ( $20\text{--}32 \text{ px}$ ) | 50.53%  |
| Medium     | area $> 1024 \text{ px}^2$ ( $> 32 \times 32 \text{ px}$ )                | 26.39%  |

### 1.3 Augmentation Setup

A YOLOv8s baseline is trained from COCO pretrained weights. Inference on the full training set yields scores via Eq. (??)–(??). Training set extended by 30% (189 synthetic images added to 536). Each synthetic image: 2–8 pasted objects, scale  $[0.9, 1.1]$ , rotation  $\pm 15^\circ$ , no blending, **same-image constraint** (objects pasted into the image they come from, to preserve photometric and contextual consistency, following Kisantal et al. [?]).

All images and objects are ranked by score and partitioned into three equal tertiles. Five conditions compared (3 seeds each):

| Condition    | Image pool     | Object pool    |
|--------------|----------------|----------------|
| Random       | Uniform random | Uniform random |
| Score High   | Top tertile    | Top tertile    |
| Score Medium | Middle tertile | Middle tertile |
| Score Low    | Bottom tertile | Bottom tertile |

## Score Validation

### 2.1 Image-Level Correlation

Table 1: Pearson correlation of  $S_{\text{img}}$  with per-image error rates.

| Dataset    | $r(\text{score, miss rate})$ | $r(\text{score, FP rate})$ |
|------------|------------------------------|----------------------------|
| MinneApple | 0.698                        | 0.688                      |
| GWHD       | 0.816                        | 0.818                      |

Moderate-to-strong correlations on both datasets. GWHD is higher because its difficulty is structured at the domain level (cleaner signal), while MinneApple has heterogeneous within-tree variation.

### 2.2 Object-Size Validity: MinneApple

Table 2: Mean  $S_{\text{obj}}$  and baseline AP by size category (MinneApple, consistent across all 3 seeds). Spearman  $r_s = -0.82$  between size rank and difficulty rank confirms strong monotonic inverse relationship.

| Category            | AP    | Mean $S_{\text{obj}}$ | Std    | In top-score tertile         |
|---------------------|-------|-----------------------|--------|------------------------------|
| Very Small (<20 px) | 0.071 | 0.1524                | 0.1433 | <b>64.39% (5138 objects)</b> |
| Small (20–32 px)    | 0.270 | 0.0755                | 0.0475 | 33.88% (2704 objects)        |
| Medium (>32 px)     | 0.493 | 0.0526                | 0.0184 | <b>1.73% (138 objects)</b>   |

**Key finding: 98.27% of high-difficulty objects are in size categories where baseline AP  $\leq 0.27$ .** Only 1.73% of the objects MIS identifies as hardest fall in the medium category where the model has genuine learning headroom (AP=0.49). This means score-guided augmentation concentrating on hard objects will predominantly paste near-undetectable instances regardless of which difficulty tier is selected.

### 2.3 Domain-Level Validity: GWHD

Table 3: GWHD domain analysis: MIS median difficulty vs. AP across 18 domains. Pearson  $r = -0.905$ , Spearman  $r_s = -0.899$ . MIS ranks domain difficulty without domain labels.

| Domain         | Median $S$ | AP    | Domain     | Median $S$ | AP    |
|----------------|------------|-------|------------|------------|-------|
| Arvalis_10     | 0.200      | 0.744 | Arvalis_8  | 0.342      | 0.546 |
| Rres_1         | 0.227      | 0.658 | Arvalis_5  | 0.342      | 0.542 |
| ETHZ_1         | 0.268      | 0.643 | Arvalis_9  | 0.372      | 0.520 |
| NMBU_2         | 0.299      | 0.639 | Arvalis_12 | 0.395      | 0.472 |
| Inrae_1        | 0.264      | 0.633 | Arvalis_3  | 0.350      | 0.470 |
| Arvalis_11     | 0.358      | 0.606 | Arvalis_7  | 0.407      | 0.463 |
| NMBU_1         | 0.335      | 0.601 | Arvalis_2  | 0.386      | 0.454 |
| Arvalis_6      | 0.289      | 0.600 | Arvalis_1  | 0.400      | 0.452 |
| ULiège-GxABT_1 | 0.380      | 0.597 | Arvalis_4  | 0.475      | 0.387 |

### 3.1 Object Size Distribution Across Conditions

Table 4: Object scale distribution per augmentation condition. *Gap* = medium-object % minus test-set target (44.70%). All gaps are negative: no condition bridges the train–test size mismatch.

| Condition        | V.Small       | Small  | Medium        | Gap       | Obj/Img |
|------------------|---------------|--------|---------------|-----------|---------|
| Test target      | 17.17%        | 38.14% | <b>44.70%</b> | 0 pp      | 37.11   |
| Baseline (train) | 23.08%        | 50.53% | 26.39%        | −18.31 pp | 42.57   |
| Random CP        | 23.52%        | 49.98% | 26.50%        | −18.20 pp | 44.03   |
| Score High       | 26.48%        | 49.59% | 23.92%        | −20.78 pp | 43.85   |
| Score Medium     | 22.56%        | 50.80% | 26.64%        | −18.06 pp | 43.23   |
| <b>Score Low</b> | <b>19.99%</b> | 48.56% | <b>31.45%</b> | −13.25 pp | 42.38   |

Score High *widens* the gap. Score Low achieves the largest shift toward test, yet still fails (see Table 4).

The maximum density increase across all conditions is **3.4%** ( $42.57 \rightarrow 44.03$  objects per image). Kisantal et al. achieved density increases of 30–60% starting from 3–5 objects per image. The mechanism that produced gains in their setting does not operate at MinneApple’s baseline density.

### 3.2 Detection Results

Table 5: 3-seed mean AP with standard deviation. All  $\Delta$ AP values are not statistically significant (ns): all differences  $< 1\sigma$  of baseline.

| Condition    | AP $\pm \sigma$                     | $\Delta$ AP | AP <sub>50</sub> | AP vs | AP s  |
|--------------|-------------------------------------|-------------|------------------|-------|-------|
| Baseline     | <b>0.353 <math>\pm</math> 0.005</b> | —           | 0.648            | 0.071 | 0.270 |
| Random CP    | 0.346 $\pm$ 0.005                   | −0.007 (ns) | 0.637            | 0.069 | 0.266 |
| Score High   | 0.350 $\pm$ 0.007                   | −0.003 (ns) | 0.645            | 0.069 | 0.271 |
| Score Medium | 0.348 $\pm$ 0.006                   | −0.005 (ns) | 0.643            | 0.072 | 0.272 |
| Score Low    | 0.348 $\pm$ 0.003                   | −0.005 (ns) | 0.643            | 0.068 | 0.271 |

AP vs = AP very small ( $< 20$  px). AP s = AP small (20–32 px).

**Score High** selects 64.39% very small objects (AP= 0.07), widens the distribution gap to 20.78 pp, and produces no improvement. Confirms that hard objects in MinneApple are near-undetectable regardless of training exposure.

**Score Low** (most important result) selects predominantly medium-sized objects, achieves the largest distribution shift (gap narrows to 13.25 pp), yet still produces no improvement. This eliminates selection strategy as the explanation for failure: even the condition best positioned to succeed cannot overcome a 3.4% density increase on a dataset already averaging 42.57 objects per image.

### 3.3 Feature Space Analysis

Feature representations extracted from YOLOv8s P5 layer, projected via PCA (PC1: 24.4%, PC2: 15.4% variance).

| Split          | PC1 centroid | PC2 centroid |
|----------------|--------------|--------------|
| Original train | +4.83        | +1.30        |
| Augmented      | +3.72        | −3.83        |
| Test           | −11.24       | −0.35        |

Augmented images remain co-located with training data ( $\Delta$ PC1  $\approx 1.1$ ) while both are separated from the test distribution by  $\Delta$ PC1  $\approx 16$ . **Copy-paste does not move the training feature distribution toward the test distribution.**

### Diagnostic Framework

The purpose of this section is to answer the question: **given MIS scores on a dataset, how do you decide whether to apply copy-paste augmentation?**

The answer is a three-criterion checklist. Each criterion is checkable from training-set inference alone, before any augmentation is run.

#### 4.1 The Three Criteria

##### Criterion 1: AP of hard objects (resolution ceiling).

Compute  $S_{\text{obj}}$  for all training objects. Find the top-difficulty tertile (hardest third by score). Identify which size/domain category those objects predominantly belong to. Look up the baseline AP for that category.

| Condition                              | Signal                                                                 | Action                        |
|----------------------------------------|------------------------------------------------------------------------|-------------------------------|
| AP of hard-object category $< 0.15$    | Hard objects near-undetectable.<br>Pasting more copies adds no signal. | <b>Do not use copy-paste.</b> |
| AP of hard-object category $\geq 0.15$ | Hard objects partially learnable.                                      | Proceed to Criterion 2.       |

*MinneApple result:*  $\text{AP}_{\text{very small}} = 0.071$ , and 64.39% of hard objects are very small. **Criterion 1 signals: do not use copy-paste.**

##### Criterion 2: Domain concentration (covariate shift).

For each image in the top-difficulty tertile, record which domain it comes from. If the majority come from domain-minority sources:

| Condition                                    | Action                                                                                    |
|----------------------------------------------|-------------------------------------------------------------------------------------------|
| Hard images concentrated in minority domains | Score-guided selection amplifies imbalance.<br><b>Do not use score-guided copy-paste.</b> |
| Hard images spread across domains            | Proceed to Criterion 3.                                                                   |

*GWHD result:* Hard images concentrate in the Arvalis minority domains (lowest AP). **Criterion 2 signals: do not use score-guided copy-paste.**

##### Criterion 3: Augmentation density headroom.

Compute: proposed density increase =  $\frac{\text{pasted objects/image} \times \text{augmentation ratio}}{\text{mean objects/image}}$ .

| Condition                    | Action                                                                                                          |
|------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Density increase $< 10\%$    | Augmentation too small to shift distribution.<br><b>Increase augmentation ratio or reduce baseline density.</b> |
| Density increase $\geq 10\%$ | Distribution shift is viable. Copy-paste may help.                                                              |

*MinneApple result:*  $(5 \text{ obj} \times 0.35)/42.57 = 4.1\%$  density increase. **Criterion 3 signals: insufficient headroom.**

#### 4.2 Summary Decision Table

Table 6: Applying the three-criterion diagnostic to both datasets. All three criteria signal failure on both. All confirmed by experiment (Table 4).

| Criterion               | MinneApple                      | GWHD                                            | Prediction  | Observed                            |
|-------------------------|---------------------------------|-------------------------------------------------|-------------|-------------------------------------|
| 1. AP of hard objects   | 0.071 $\rightarrow$ <b>Fail</b> | Mixed (domain-dependent)                        | Fail        | Fail $\checkmark$                   |
| 2. Domain concentration | N/A (single domain)             | Minority-concentrated $\rightarrow$ <b>Fail</b> | Fail        | Fail $\checkmark$                   |
| 3. Density headroom     | 3.4% $\rightarrow$ <b>Fail</b>  | Similar $\rightarrow$ <b>Fail</b>               | Fail        | Fail $\checkmark$                   |
| <b>Overall</b>          |                                 |                                                 | <b>Fail</b> | <b>Fail <math>\checkmark</math></b> |

#### 4.3 When Copy-Paste Is Likely to Help

The framework also implies the positive conditions — the dataset properties where copy-paste is worth attempting:

- AP of hard objects  $\geq 0.15$  (learnable, not yet saturated),
- High-difficulty images spread across domains (no concentration),
- Density increase  $\geq 10\%$  (sparse dataset, augmentation moves the distribution).

These conditions are consistent with Kisantal et al.’s success on COCO (sparse images, 30–60% density increase, large learnable objects). On MinneApple and GWHD, none of these conditions hold, which explains the uniform failure.

#### 4.4 Limitations of the Framework

The three criteria were derived from the same datasets used to validate them, so this is a retrospective characterization, not a prospective prediction tested on an unseen dataset. The thresholds (AP  $< 0.15$ , density  $< 10\%$ ) are operationally reasonable but not formally derived. A prospective test on a new dataset would strengthen the framework substantially.

## Proposed Paper Structure (GCCE, 2 pages)

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| Section         | ≈Words | Content and main point                                                                                                                                     |
|-----------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Abstract        | 100    | MIS scorer. 5 conditions. 64.39% hard = very small. 18.31 pp mismatch. Decision rule. No positive AP claimed.                                              |
| I. Intro        | 160    | Copy-paste fails on agricultural data. No diagnostic exists. Two contributions: (1) MIS scorer, (2) structural diagnostic study.                           |
| II. Related     | 100    | OHEM/Focal Loss (training-time, different purpose). Kisantal (anchor-based, mechanism does not transfer to YOLOv8s). MIS: post-hoc, architecture-agnostic. |
| III. Method     | 200    | Eq. 1–2 with correct $\tau$ values. Five conditions. Same-image justification. Custom thresholds with justification.                                       |
| IV. Diagnostic  | 200    | Table: Pearson. Table: size rankings + 64.39% finding. GWHD $r = -0.905$ . 18 pp mismatch. Three-criterion decision table.                                 |
| V. Augmentation | 250    | Combined AP + distribution table. Score Low analysis (most informative result). PCA figure. All deltas marked (ns).                                        |
| VI. Conclusion  | 100    | Structural failure confirmed. Decision rule in 3 steps. Future: generative augmentation targeting the learnable zone.                                      |

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## Known Errors to Fix Before Submission

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1. **Verify GWHD split.** Confirm GWHD used a domain-based split (not random image-level). If contaminated, retrain or limit claims to MinneApple only.
  2. **Rename “boundary sampling.”** The implementation is tertile filtering, not class-boundary sampling in the ML sense. Use “difficulty-tier selection” throughout.
  3. **Do not write “predicts.”** The framework is retrospectively validated on the same datasets used to derive it. Write “identifies structural conditions consistent with failure” rather than “predicts failure.”
  4. **Change “strongly positive”** to “moderate-to-strong ( $r = 0.698$ – $0.818$ )” for Pearson correlations.  $r = 0.698$  is moderate, not strong.
  5. **Fix all typos** from original draft before submission.

## References

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